



INSECT-SCALE AUTONOMY

Reliable autonomy when sensing, compute, power, actuation, and mass are all scarce.

SUBMISSION DEADLINE

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**3–4 PAGE PAPERS / 1–2 PAGE POSTER
ABSTRACTS**

Three tightly coupled research fronts

We invite research that connects algorithms, bodies, and hardware for reliable small-scale autonomy.

A

RESOURCE-AWARE PERCEPTION & LEARNING

Extract task-relevant information under severe energy, memory, bandwidth, and latency limits.

B

MECHANICS, ACTUATION, DESIGN & CONTROL

Co-design morphology, compliant mechanics, actuation, sensing, and feedback control.

C

ON-BOARD COMPUTE & EDGE AI

Deploy estimation, planning, learning, and control within strict power, thermal, and mass budgets.

CONFIRMED SPEAKERS

Yiannis Aloimonos
University of Maryland

Vijay Janapa Reddi
Harvard University

Frances Chance
Sandia National Laboratories

Geoffrey Barrows
CentEye, Inc.

Elia Cereda
IDSIA, USI–SUPSI

POST-WORKSHOP OUTCOME

Join a community perspective paper defining shared bottlenecks, evaluation needs, and a research roadmap for insect-scale autonomy.

SUBMIT YOUR WORK

insect-autonomy.github.io

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